meet the quantum

In its simplest form, quantum physics is the study of matter and radiation at an atomic level, where things work very differently from "our" world. A good example of this difference can be seen in the two broad kinds of phenomena that we are used to: particles and waves. Particles exist in one place at a time. Waves, such as sound waves, are spread out in space. At the atomic level, this distinction does not apply. Electrons, which are thought of as particles, can behave like waves. Similarly, in the case of light, which used to be thought of as occurring in waves, some behavior can only be explained if light exists in the form of particles, known as photons. This wave-particle double-nature, or "duality," can only be accounted for by quantum physics. Other new quantum phenomena include the discreteness of energy, quantum tunneling, the uncertainty principle, and the "spin" of a subatomic particle, all of which we are about to explore.

making waves

Many people are familiar with the idea that light travels through space in the form of a wave, like ripples on a pond. This was established at the beginning of the 1800s. But in quantum physics, we have to come to terms with the fact that things like electrons, which we like to think of as particles, also travel as waves. A key property of waves is that, unlike particles, when they pass through a small hole, they bend to the sides, and spread out, in a process called diffraction.